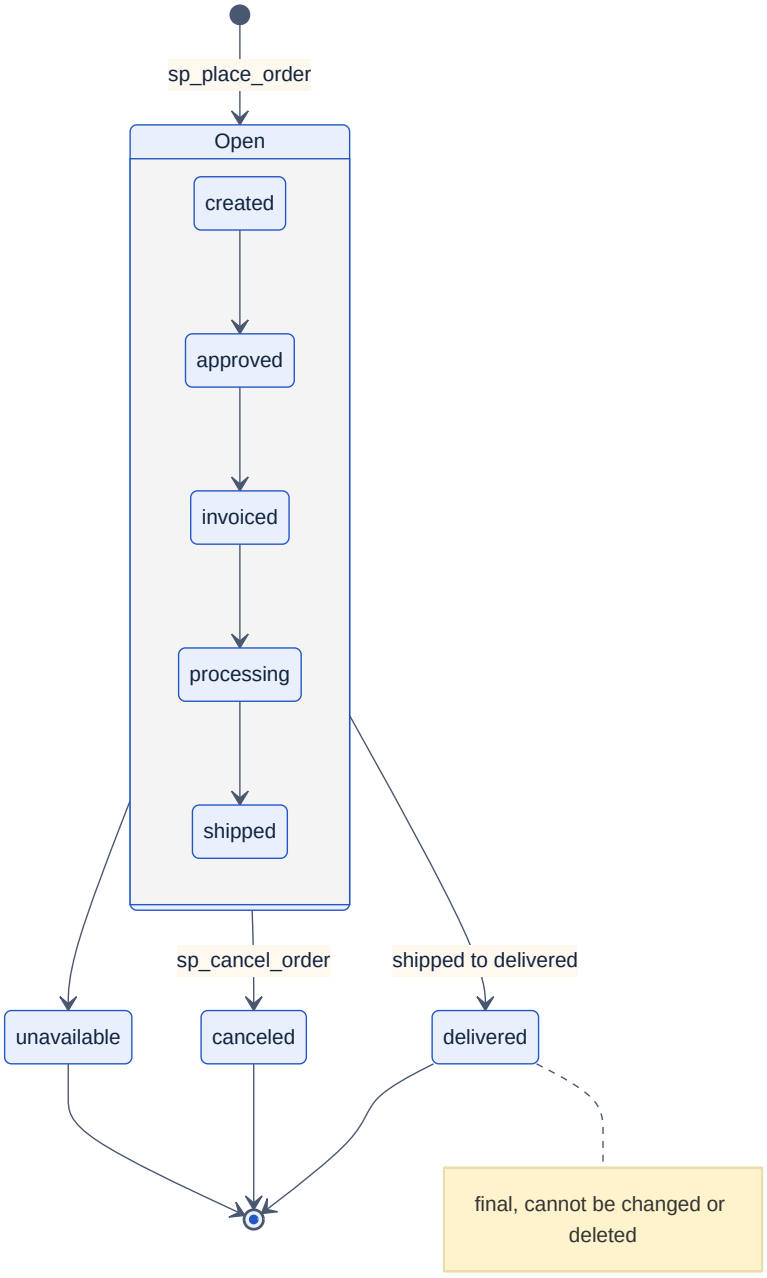
UML Diagrams

Project: E-Commerce & Order Management System (Olist) · **Group:** TarushTanishkTanishq
Version: 1.0 (mid-semester) · Diagrams: state machine, sequence (×2), activity, class

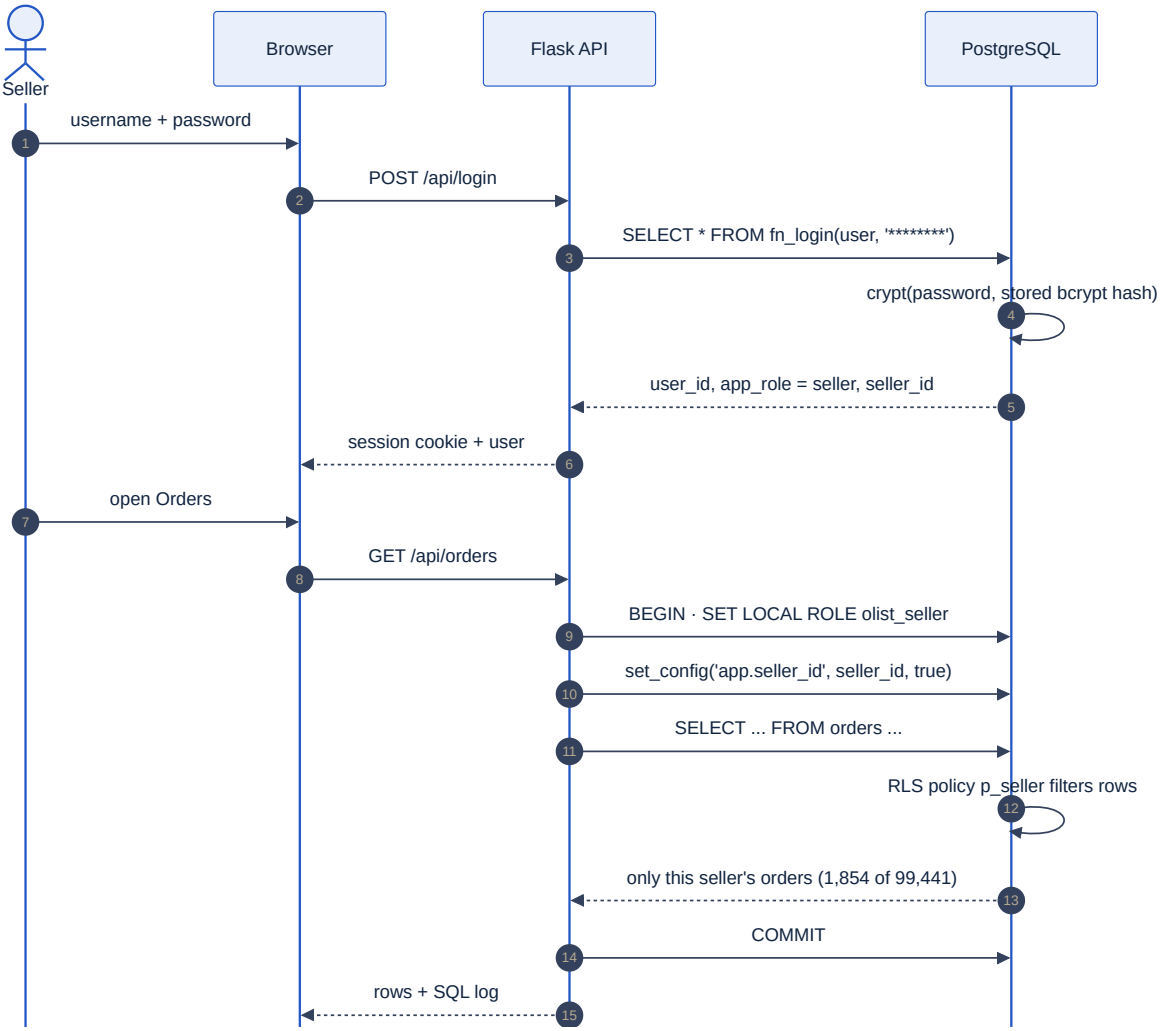
1. State machine: order status

Enforced by trigger `trg_validate_status_transition` (forward moves only) and audited by `trg_order_status_log`.

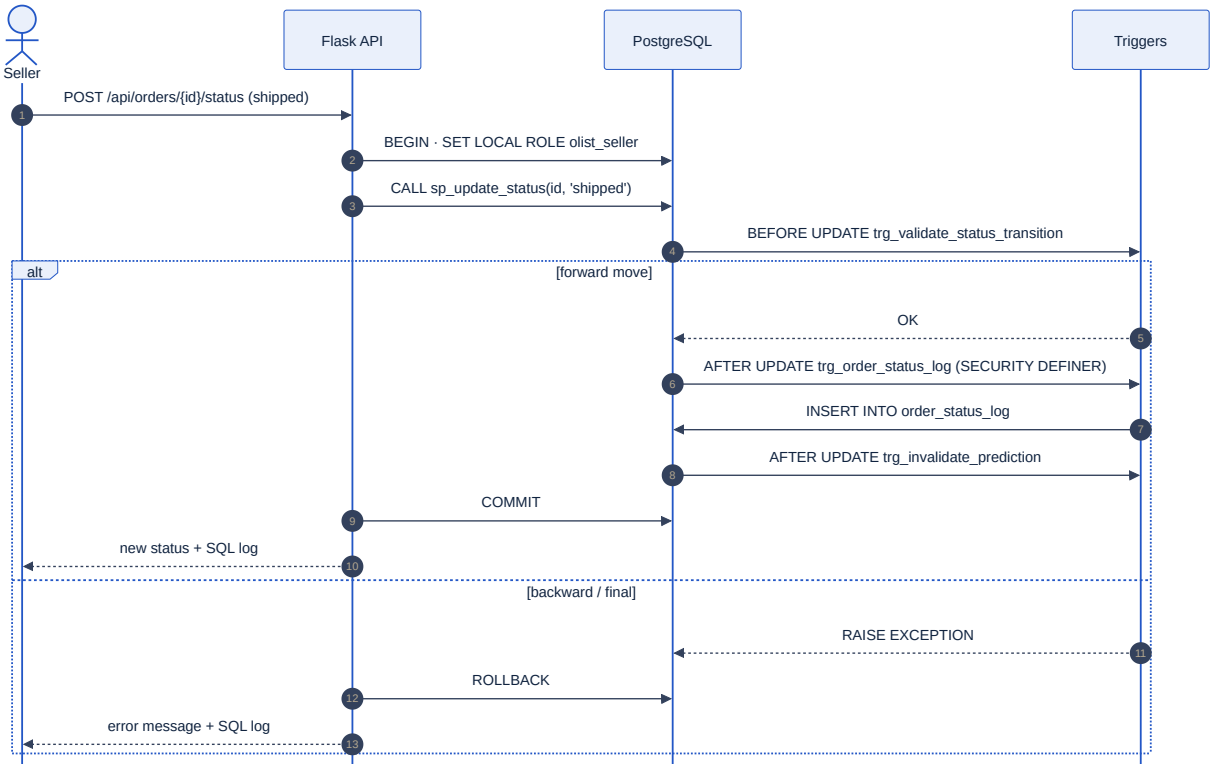


Inside *Open*, any forward move is allowed, including skipping states (e.g. `approved → shipped`, test TC-07); `canceled` and `unavailable` can be reached from any open state. Backward moves and changes after a final state raise an error (TC-06). Every successful change writes one row to `order_status_log` (TC-08).

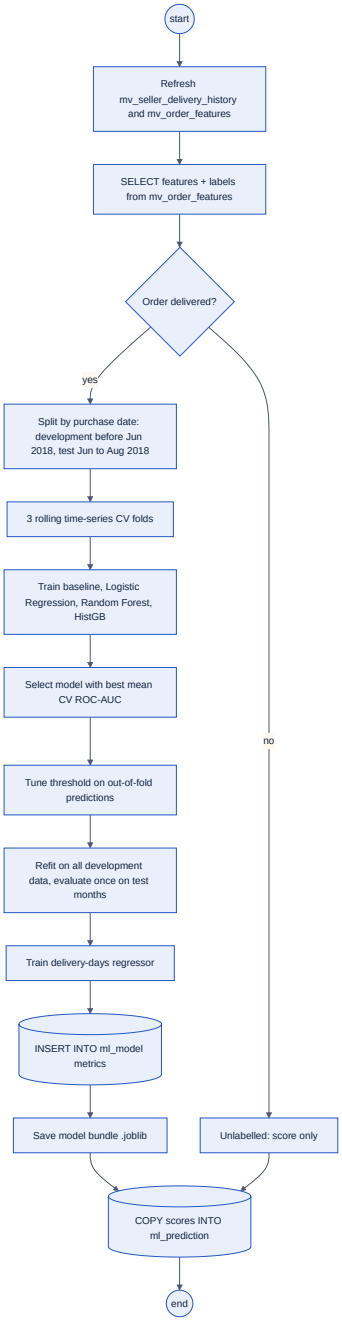
2. Sequence diagram: login and role-based access



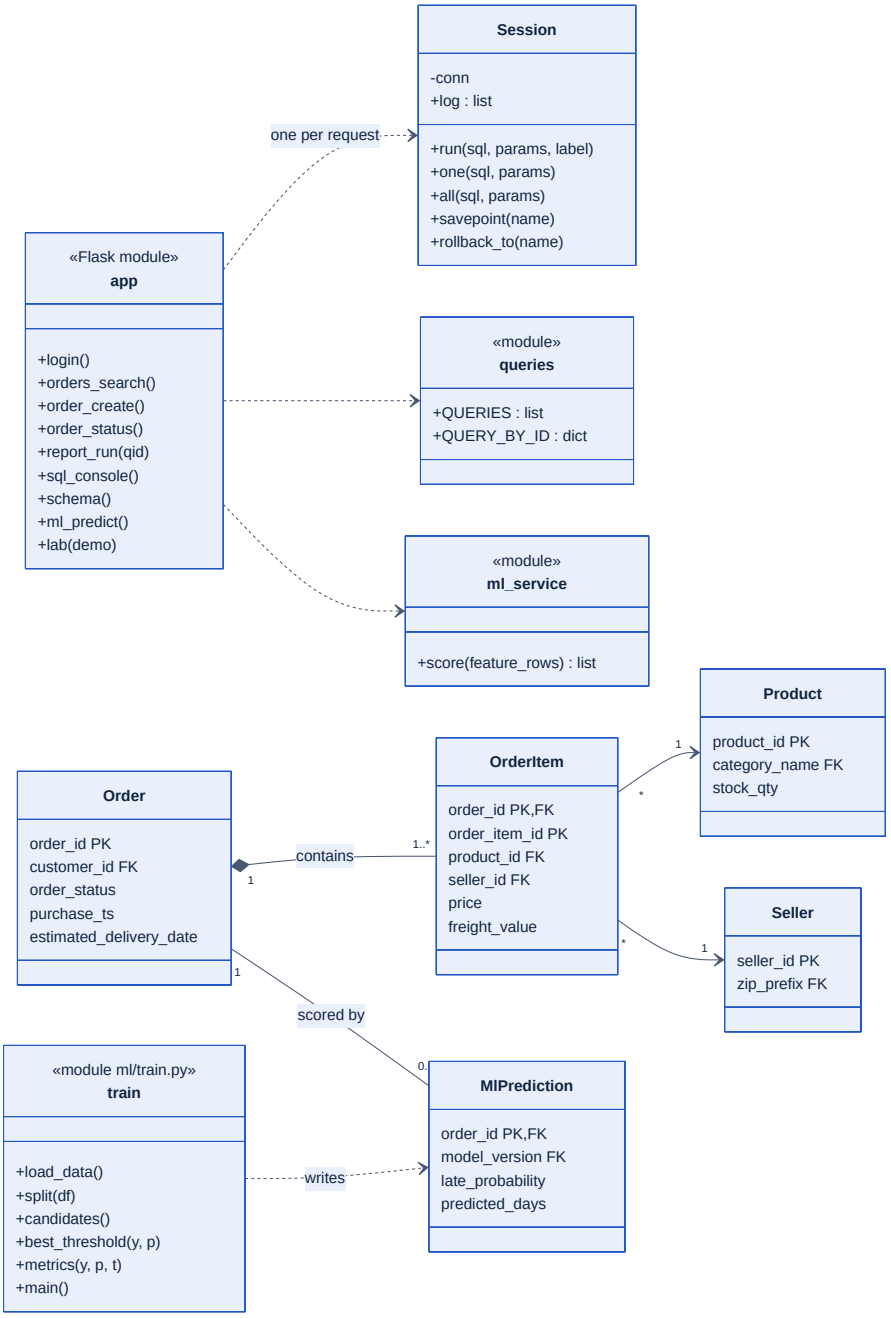
3. Sequence diagram: seller updates an order status



4. Activity diagram: ML training pipeline (`ml/train.py`)



5. Class diagram: application modules and main domain classes



`Session` (in `backend/db.py`, created by the `session()` context manager) opens one transaction per request, sets the role, binds parameters on the client side and records every statement for the SQL panel. The domain classes mirror the main tables; the full data model is in *Database_Design.pdf*.